Nathan Espejo

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Education

Western University — London, ON
Bachelor of Engineering Science in Software Engineering, Dean's Honour List (2024, 2025) — *Aug 2024 - May 2028*

Experience

VP of Data/ML — May 2026 - Aug 2026
3D Western — *London, ON*

- Co-led two data developers on the print-safety dataset and benchmark, tracking work on a GitHub project board.
- Built a labelled dataset from 4,620 scraped 3D models, rendered in Blender with labels checked by a VLM.
- Benchmarked small vision models as a binary safe-or-flag gate: 84.2% best agreement with VLM-checked labels.
- On the data team since Sep 2025, writing the render scripts and working on the 3D model scraping and the STL rendering pipeline.

Software Engineering Intern — May 2026 - Aug 2026
TeraGo — *Markham, ON*

- Built a prototype pipeline that uses a locally hosted LLM (Ollama) to turn unstructured emails into structured tickets, working end to end.
- Built a Python RF channel-planning tool with a command-line and web interface that shortlists clear frequencies for fixed-wireless links, with an automated validation suite.
- Deployed an internal RAG assistant that answers engineers' technical questions from company documentation.
- Researched and proposed a self-hosted LLM platform integrated with internal tools.

Projects

Campus | *Python, FastAPI, SQLite, React, TypeScript, Docker* — Aug 2026 - Present

- Built an AI agent that reads all six of my courses, syncing them from the LMS into SQLite with Playwright and running on any OpenAI-compatible endpoint, local or hosted.
- Designed search over course files that is lexical by default with optional embeddings and reranking, and added a lexical SQL prefilter for exact phrases that scored below the semantic cutoff.
- Wrote a 20-tool agent harness with an audit log for every AI write and a bounded tool loop, with 403 backend tests in CI.
- Use it across six courses for deadline lookups, studying from course slides, and reviewing drafts against briefs.

Homelab & NixOS | *NixOS, Nix flakes, Docker, Tailscale, Caddy* — May 2026 - Present

- Defined a laptop and home server in a single Nix/NixOS repo, so every change can be reviewed and rolled back.
- Self-host passwords, photos, files, a git remote and AI services behind Caddy and Tailscale, with sops-nix secrets.
- Set up daily NAS backups, uptime alerts, intrusion detection with CrowdSec, and split-horizon DNS.
- Run a Hermes Agent per machine that edits only the config, with my approval required on the server.

AWARE (Hack the 6ix, solo) | *Python, YOLOv8, YAMNet, ONNX, Qualcomm QNN* — Jul 2026

- Built a home-awareness device running YOLOv8 and YAMNet on an $80 Arduino UNO Q, no cloud inference.
- Used an on-device LLM to compile commands like "when glass breaks after 10pm alert me" into deterministic rules, checked every 500 ms with no LLM in the loop.

Hackathon Awards

8 awards in 22 hackathons: NSBEHacks 3rd place, 1st for Cohere (Careerly) | Hack Western Best AI Application with Cloudflare (Auralis) | TerraHacks Best Use of MongoDB (Lumen) | Spark 2026 3rd (CareBridge, contributor)

Technical Skills

Languages: Python, TypeScript/JavaScript, Java, C#, C++, Kotlin, SQL, Bash
AI/ML: LLM agents, tool calling, RAG, embeddings and reranking, MCP, VLMs, model evaluation, llama.cpp, Ollama, PyTorch, scikit-learn, pandas, YOLOv8, ONNX, MediaPipe
Web & Data: React, Next.js, Tailwind, Three.js, FastAPI, Node.js, PostgreSQL (Supabase), MongoDB
Infra: Nix/NixOS (flakes), Linux, Docker, GitHub Actions, Tailscale, Caddy, Git
Hardware & 3D: Arduino, ESP32, Raspberry Pi, Unity (XR), Blender, Onshape, 3D printing